Data path

Public captures

CICIoT2023

1,045,666 unique flows, 43 feat., 8 cls

NF-UNSW-NB15-v3

2,242,931 flows, 47 feat., 10 cls

Preprocessing

drop identifiers and timestamps

CIC: remove exact duplicates first

signed-log map; alternatives in V-F

Section IV-A

Split

CIC: stratified, 3 partitions

NF: train $<$ val $<$ test in time

no test row used for selection

Section IV-A

Training

K = 2..6 and matched controls

13 neural configs, 10 seeds each

13 linear, additive, tree baselines

Sections IV-B, IV-D, IV-E

Evaluation

held-out flows: 313,700 + 672,880

macro PR-AUC, per class, recall at fixed FPR

exact paired permutation, Holm/Bonferroni

Sections IV-C, IV-D, V-B, V-I

Audit path — available because the logit is exactly decomposable

Trained detector

weights W_k, V_k, $\widehat{s}_k$, b

Section III

Per-degree contributions

$\varphi_k = V_k \bar{u}_k$ for every flow

three-level exactness checks

magnitude and margin shares; removal

Sections V-A, V-C, V-D, V-E

exactness at three levels; how much of the logit and of the decision margin each degree carries; what is lost when a degree is removed

Coefficient expansion

closed form of Section III-E

paths folded onto monomials

arity against a null; seed stability

Section V-G

largest learned coefficients per class, their stability across retraining, and interaction arity against combinatorial and population references

Perturbation test

marginal, joint, stratified replacement

999 random controls per class

recall, prediction flips, margin

Section V-H

whether the features named by the coefficients affect detection more than random feature sets under dependency-preserving replacement